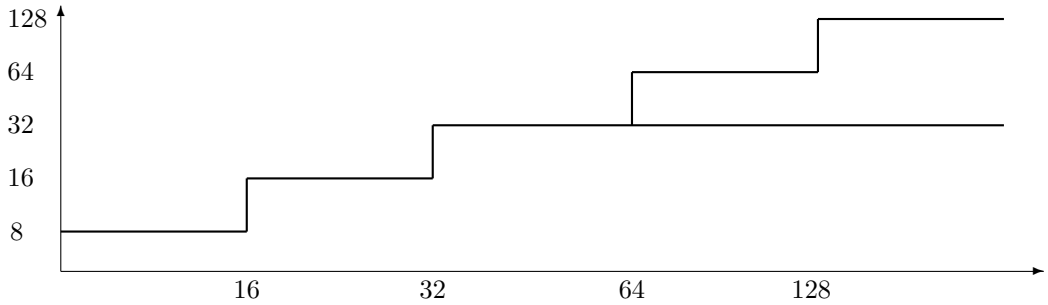
Independent GLV size-dependent cap proof review

2026-11-07. Nineteen new named Lean theorem types describe the pinned C selector: cap 32 below $lg = 40$, cap 128 from $lg = 40$, divisor 12, start value 8, and four possible doublings. The new selector equals the previous one below 40 bits and cannot choose a smaller table for the same numerator and log factor.

Budget b	cap 32	cap 128
$b < 16$	8	8
$16 \le b < 32$	16	16
$32 \le b < 64$	32	32
$64 \le b < 128$	32	64
$128 \le b$	32	128

Machine checks. Fresh pinned Lean 4.34.1 compilation passed. All 27 declaration axiom prints used only `propext` and `Quot.sound`; the source file has no `sorry`, custom axiom, `unsafe`, or `native_decide`. Seven supplemental hashes passed. The live C-source binding checker passed and rejected five changed-selector controls. Blind arithmetic and separate source-fidelity reviews agree on the conditional selector result. **Inputs and boundary.** Target main is `702b6804`; current C SHA-256 is `e1d624e5...f63`. The C path yields a positive log factor, but the proof does not establish the external `c.m>0` prerequisite for C division. No performance or solver-correctness trial was rerun.



Blue/low branch in the companion SVG is cap 32; orange/high branch is cap 128. The figure is a proved selector diagram, not a performance plot. **Correction needed.** `LEAN_PROOF.md` says the loop doubles only 8 and 16. The cap-128 path can also double 32 and 64. This narrative issue does not change the theorem types. See the companion Markdown, reviewer attestations, and machine receipts for full evidence.